

Dear Editors,

Following the editorial recommendation from Nature Methods, we are pleased to submit our manuscript, “A distribution-aware framework for prioritizing heterogeneous disease-relevant responses in single-cell perturbation data,” for consideration as an Article in Communications Medicine.

Single-cell perturbation screens and patient-derived single-cell datasets are increasingly used to nominate disease mechanisms and therapeutic targets, but many analyses still prioritize average expression changes. This can under-rank responses that occur in only a subset of cells, a pattern that is common in heterogeneous malignancies, immune responses, inflammatory tissues, and CRISPRa perturbation systems. PGAA addresses this problem by providing a distribution-aware framework for ranking perturbation-associated transcriptional responses, combining a Wasserstein statistic for full expression-distribution shifts with a persistent-homology statistic for responder-associated expression-shape changes.

We believe the manuscript fits Communications Medicine because it presents a computational method and software resource with translational relevance to disease-focused single-cell perturbation analysis. In five disease-relevant observational scRNA-seq datasets, PGAA recovers known marker programs, including B-cell receptor signaling in chronic lymphocytic leukemia. We interpret these analyses conservatively as marker recovery rather than causal validation and now present them as a main result rather than as supplementary evidence. We then benchmark the method in two experimental Perturb-seq datasets, foregrounding the Adamson 2016 unfolded-protein-response CRISPRi benchmark where the Wasserstein statistic outperforms Wilcoxon, t-test, and MAST in a small independent setting. The Norman 2019 CEBPE CRISPRa analysis is presented as a narrow persistence-ranking example rather than broad target-program recovery. Calibration and simulation analyses define the method’s operating regimes and limitations.

The manuscript has been revised specifically for Communications Medicine. We have foregrounded the medical problem of heterogeneous disease-relevant transcriptional responses, clarified the distinction between observational marker recovery and experimental perturbation validation, and avoided claiming new causal disease mechanisms or clinical biomarkers from public observational datasets. Instead, the manuscript positions PGAA as a transparent prioritization layer for heterogeneous responses that can be followed up in disease models, patient-derived systems, or therapeutic perturbation experiments.

The software is implemented in Python and R. A public code-only repository is available at <https://github.com/xutaoguo55/pgaa>, with release v0.1.0-code archived at Zenodo (DOI: <https://doi.org/10.5281/zenodo.20681141>) and by Software Heritage (SWHID: [swh:1:snp:5b1b2cc9ce32298968e00f69e1af5ff8aed8889f](https://swhid.org/urn:uuid:5b1b2cc9ce32298968e00f69e1af5ff8aed8889f)). Manuscript-specific source-data tables, figure-reproduction scripts, runtime documentation, and PDF build files are provided in the submitted supple-

mentary software archive rather than in the public code-only repository. All authors have approved the manuscript. We declare no competing interests.

Thank you for considering our manuscript.

Sincerely,

Xutao Guo, on behalf of all authors

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